3. Computer Memory

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Content

1. Introduction to Computer Memory

Computer memory is any physical device capable of storing information temporarily or permanently. It is a critical component without which a computer would not be able to function.

Key Functions:

- **Store Data and Instructions**: Holds the data to be processed and the instructions (programs) for processing it.
- **Provide Access to CPU:** Supplies the Central Processing Unit (CPU) with the required data and instructions at high speed.

2.Types of Computer Memory

Computer memory can be classified based on various characteristics like volatility, access method, storage capacity, and physical location. Here's a comprehensive classification:

1. Based on volatility
2. Based on access method
3. Primary memory
4. Secondary memory
5. Cache memory
6. CPU Register

2.Types of Computer Memory

Computer memory can be classified based on various characteristics like volatility, access method, storage capacity, and physical location. Here's a comprehensive classification:

1. Based on Volatility

1.1 Volatile Memory

Loses stored data when power is turned off

- RAM (Random Access Memory)
  - DRAM (Dynamic RAM)
  - SRAM (Static RAM)
- Cache Memory (L1, L2, L3)
- CPU Registers

1.2 Non-Volatile Memory

Retains data even when power is turned off

- ROM (Read-Only Memory)
  - PROM(Programmable Read Only Memory)
  - EPROM(Erasable Programmable Read Only Memory)
  - EEPROM(Electrically Erasable Programmable Read Only Memory)
- Flash Memory
- Secondary Storage
  - HDD (Hard Disk Drives)
  - SSD (Solid State Drives)
- Optical Storage (CD, DVD, Blu-ray)
- Magnetic Tape

2.Types of Computer Memory

2. Based on Access Methods

2.1 Random Access Memory (RAM)

Any storage location can be accessed directly in any order

- DRAM - Main system memory
- SRAM - Cache memory
- Flash Memory - SSDs, USB drives

2.2 Sequential Access Memory

Data must be accessed in a predetermined sequence

- Magnetic Tape
- Punched Cards
- Cassette Tapes

2.3 Direct Access Memory

Combination of random and sequential access

- Hard Disk Drives (HDD)
- Optical Discs (CD/DVD)

2.Types of Computer Memory

3. Primary Memory (Main Memory)

Primary memory is directly accessed by the CPU. It is fast but has limited capacity.

3.1. RAM (Random Access Memory)

- Temporary, volatile memory
- Stores data and instructions currently in use
- Data is lost when power is off

Types of RAM:

- **DRAM (Dynamic RAM):** Slower, cheaper, used in main memory
- **SRAM (Static RAM):** Faster, expensive, used in cache

3.2. ROM (Read Only Memory)

- Permanent, non-volatile memory
- Stores essential instructions like BIOS
- Cannot be easily modified
- Data written during manufacturing

Types of ROM:

- **PROM** – Programmable once
- **EPROM** – Erasable using UV light
- **EEPROM** – Electrically erasable

2.Types of Computer Memory

4. Secondary Memory (External Memory)

Secondary memory stores data and programs permanently. Characteristics: ***Non-volatile, High storage capacity, Slower than primary memory***

4.1 Magnetic Storage

Hard Disk Drive (HDD)

- Uses magnetic platters
- Mechanical moving parts
- High capacity, low cost
- Sequential access

Magnetic Tape

- Very high capacity
- Very slow access
- Used for backups and archives

4.2 Solid State Storage

SSD (Solid State Drive)

- Uses NAND flash memory
- No moving parts
- Faster than HDD
- More expensive per GB

Flash Memory Types:

- NAND Flash - Used in SSDs, USB drives
- NOR Flash - Used for firmware storage

4.3 Optical Storage

- CD (700 MB)
- DVD (4.7 GB - 17 GB)
- Blu-ray (25 GB - 128 GB)

2.Types of Computer Memory

5. Cache Memory (L1, L2, L3)

The cache is a small, very fast memory that sits between the CPU and the main memory (RAM). It stores copies of frequently used data from main memory.

- Location:
  - L1 Cache: Inside each CPU core.
  - L2 Cache: Also inside each core (or sometimes shared between cores).
  - L3 Cache: Shared among all CPU cores on a chip.
- Capacity: L1 (64-512 KB), L2 (256 KB - 2 MB), L3 (4 - 64 MB).
- Speed: Very fast. L1 is the fastest (a few cycles), then L2, then L3.
- Cost: Very expensive.
- Volatility: Volatile.
- Function: Reduce the average time to access data from the main memory (RAM). This is based on the principle of locality (temporal and spatial).

2.Types of Computer Memory

6. CPU Registers

Registers are very small, ultra-fast memory units inside the CPU. Registers provide the fastest data access but store very small amounts of information.

- **Location**: Inside the CPU core.
- **Capacity**: Very small (a few KB total). Each register holds a single piece of data (e.g., 32 or 64 bits).
- **Speed**: Fastest. Access time is 1 CPU cycle. Operations are performed directly on register contents.
- **Cost**: Most expensive.
- **Volatility**: Volatile.
- **Function**: Hold the operands and results of the current instructions being executed by the CPU.

2.Types of Computer Memory

Memory Type	Volatility	Speed	Cost	Typical Use
CPU Registers	Volatile	Fastest	Highest	CPU operations
SRAM Cache	Volatile	Very Fast	Very High	CPU cache
DRAM	Volatile	Fast	Medium	Main memory (RAM)
Flash Memory	Non-Volatile	Medium	Medium-High	SSDs, USB drives
HDD	Non-Volatile	Slow	Low	Mass storage
Magnetic Tape	Non-Volatile	Very Slow	Very Low	Archives, backups
ROM	Non-Volatile	Fast	Low	Firmware, BIOS

3.Memory Hierarchy

The Memory Hierarchy is a concept that organizes different types of memory in a computer based on their speed, cost, and capacity.

The Fundamental Trade-off:

- Faster memory is more expensive per byte.
- Cheaper memory allows for larger capacity.

To create a system that is both high-performance and cost-effective, computers use a pyramid-like structure of memory types.

The Memory Hierarchy Pyramid

(Top = Fastest, Smallest, Most Expensive |

Bottom = Slowest, Largest, Cheapest)

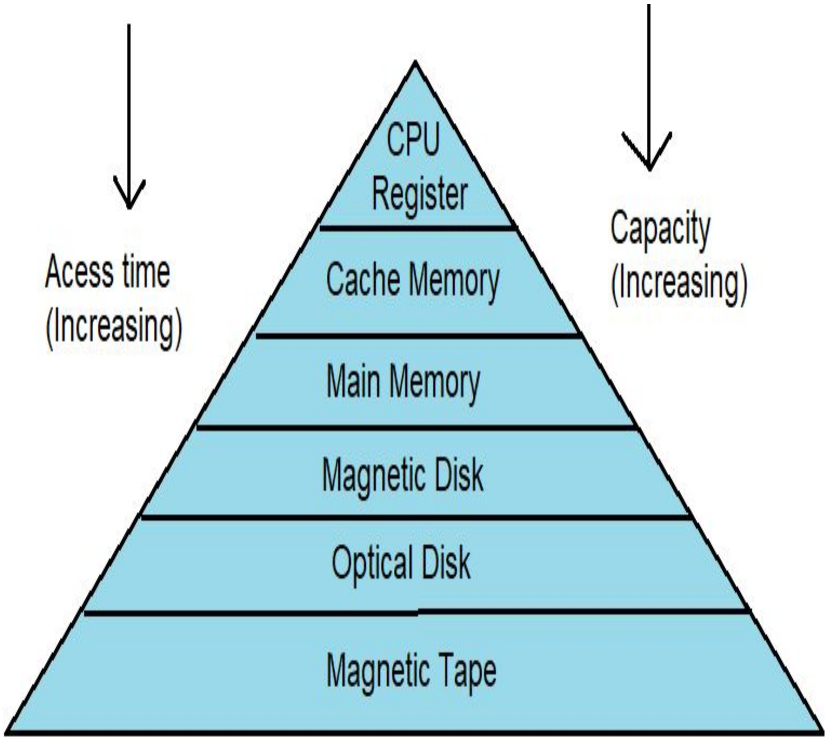

Fig:- Memory Hierarchy

4.How Data Moves Through the Hierarchy: A Simplified Workflow

1. The CPU needs a piece of data.
2. It first checks the fastest, closest memory: the L1 Cache.
3. If the data is found (a cache hit), it is used immediately.
4. If not found (a cache miss), it checks the L2 Cache.
5. If it misses again, it checks the L3 Cache.
6. If the data is not in any cache, the CPU accesses the Main Memory (RAM).
7. The required data is fetched from RAM and a copy is placed into all levels of cache (based on the algorithm) for future access.
8. If the data is not in RAM, it is loaded from Secondary Storage (SSD/HDD) into RAM first, and then the process repeats.

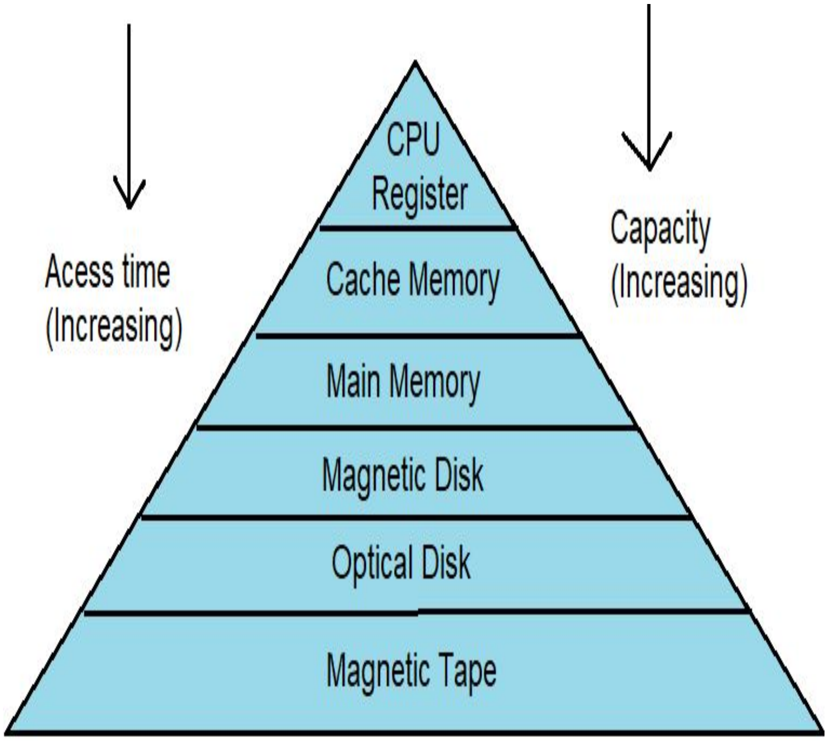

Fig:- Memory Hierarchy

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Chapter 3